

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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Anonymous submission

ABSTRACT

We quantify how physical scale and population size shape the emergence of complex behaviors in open-ended ecological environments. In our setting, agents are unsupervised and have no explicit rewards or learning objectives but instead evolve over time according to reproduction, mutation, and selection. As they act, agents also shape their environment and the population around them in an ongoing dynamic ecology. Our goal is not to optimize a single high-performance policy, but instead to measure how behaviors emerge and evolve across large populations due to natural competition and environmental pressures. Using modern hardware together with a new multi-agent simulator, we scale the environment and population well beyond prior attempts, sustaining populations of over 60,000 agents, each with its own evolved neural network policy. Across a controlled sweep of five world sizes spanning 64×64 to 1024×1024 grids, we identify emergent behaviors such as long-range resource extraction, vision-based foraging, and predation that arise under competitive and survival pressures, and we measure precisely when and how reliably each behavior appears. Comparing matched populations that differ only in a single sensor or in world size, we find sharp, consistent gaps: several behaviors emerge only in sufficiently large environments and populations, and larger scales markedly increase the stability and consistency of these emergent behaviors. While there is a rich history of research in evolutionary settings, the clear scaling trends we report on modern hardware provide promising new directions to explore ecology as an instrument of machine learning in an era of increasingly abundant computational resources and efficient machine frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been studied across a wide variety of biological disciplines, including evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Models of these emergent dynamics range from theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) to the mathematical analyses of population genetics (Fisher, 1930; Wright, 1931).

In these fields, decades of mathematical modeling, laboratory experiments, and field work have tremendously improved our understanding of the mechanisms of evolution in nature. Unfortunately, there are many obstacles to studying emergent behavior in the natural world. As ecosystems become larger and more complex, they are more difficult to control and measure. Even where possible, running controlled experiments in large-scale natural settings risks damaging or displacing wild populations.

In an effort to better understand the emergence of complex behavior due to competitive and environmental pressures, we study the open-ended evolution of neural network policies in large-scale ecological simulations. In this setting, agents do not have a specified objective or reward signal but instead collect resources to survive and reproduce according to the dynamics of the environment. Policy changes occur only via mutation as parents pass their policies on to their children. With the rapid growth in computational resources available to researchers, our goal is to study open-ended evolutionary dynamics in populations of these simple digital organisms at scales not previously pos-

sible, in the hopes of providing new tools for investigating the effects of environmental scale and population size on behavioral emergence.

To study this, we use a new JAX-based simulation environment that allows rapid evaluation of large grid worlds. This lets us hold every agent and environment rule fixed while varying scale across five grid sizes, from 64×64 up to 1024×1024 : our larger experiments contain over 60,000 concurrent agents in a physical area exceeding 1,000,000 grid cells. Agents in this environment must navigate to find food and other resources in order to survive. Agents can also reproduce by collecting enough resources to make a mutated copy of themselves. A 3D visualization of this environment can be found in Figure 1.

Our goal is not only to better understand ecological phenomena through simulation, but to also study ecology as a mechanism for producing machine intelligence. Recent years have demonstrated that in AI, scale doesn't merely improve models—it fundamentally transforms their capabilities (Wei et al., 2022a). Yet despite this insight, many attempts at embodied learning have been often been confined to relatively small environments with low populations. Our work explores what capabilities emerge when we scale not just a single model but an entire ecosystem, and it treats scale as the primary experimental variable to be measured rather than a fixed backdrop. Just as reasoning emerges in individual models only beyond a certain size threshold (Wei et al., 2022b), we seek to quantify which strategies evolve within populations only once environmental size and complexity cross a sufficient threshold, and how sharply the reliability of those strategies improves with scale.

Through controlled experiments that isolate one factor at a time, we quantify multiple instances where sensor configurations and our large environmental scales strongly impact the behaviors that emerge from simulation, which can lead to long-term ecological consequences. For example, holding all else fixed and toggling a single compass sensor, agents adapt to conduct long-range resource gathering expeditions: this “mining” behavior emerges in every seed at and above the 256×256 scale for compass agents, but is essentially absent in matched compass-free populations, which instead go extinct in all four seeds at every scale up to 512×512 . A survival analysis over 192 runs sharpens the comparison: enlarging the world from 128×128 to 512×512 cuts the extinction hazard by roughly $25\times$ and speeds mining discovery by about $6\times$, while enabling the compass accelerates mining emergence by roughly $35\times$. We also show that agents equipped with vision evolve markedly better foraging and attacking behavior than blind agents, sustaining populations roughly $2\text{--}3\times$ larger and attacking far less often but with a kill rate near 75%; here too the advantage widens and stabilizes at larger scales. Taken together, these consistent, large-margin trends demonstrate that the scale and complexity enabled by modern hardware accelerators offer promising new approaches to study how evolution in ecological environments can produce rich new behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field's inception. Conway's Game of Life (Games, 1970) simulates populations through simple local rules. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown) simulate life as self-replicating programs that evolve through mutations. Like Conway, Reynolds (1987) used local rules of motion to produce emergent flocking and schooling behaviors.

Researchers have used these principles to evolve the locomotion morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022; Muga & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods have evolved neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). In contrast to these approaches, we do not attempt to produce agents that can better adapt to their environment using some reward or objective signal, but instead seek to understand what behaviors naturally emerge as a result of ecological pressure and understand the impacts of environmental scale and complexity on these behaviors.

Evolution has also provided inspiration for the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) apply the principles of evolution and natural selection to optimization. Other similar works use evolutionary

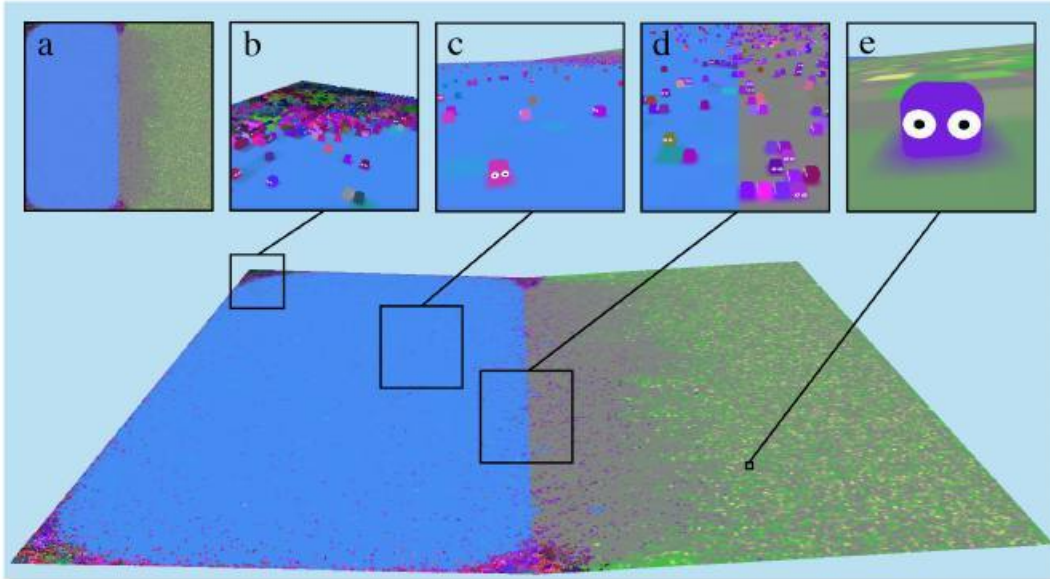


Figure 1: A 3D visualization of a 512×512 map in our environment. Insert (a) shows a top down view of the entire map, (b) shows a group of agents clustered in the corner, (c) shows agents in the water, (d) shows agents travelling back and forth from the land and (e) shows a close up of a single agent gathering resources.

methods to encourage novelty within the resulting solutions (Lehman & Stanley 2011, Cully et al. 2015). Other works have used these approaches to optimize neural networks for tasks like Atari games and Mujoco locomotion tasks (Salimans et al. 2017, Such et al. 2017, Mania et al. 2018). While our work employs evolutionary mechanisms to update the agents, our focus is on studying emergent behavior rather than developing a novel optimization algorithm or using an existing one to accomplish a fixed objective.

There is also a rich history of work which takes biological inspiration to develop organisms within an environment. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). Some recent works use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al. 2020, Robinson 2021, Heinemann 2021, Randazzo & Mordvintsev 2023, Caussan 2025). Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to investigate the emergence of language and communication between agents. The emergence of behavior due to autocurricula (Leibo et al. 2019), the idea that populations of agents provide an increasingly challenging learning landscape for each other, has also been explored in environments with smaller populations than ours (Leibo et al. 2018, Team et al. 2021). Additionally, Wang et al. (2019) and Aki et al. (2024) take the reverse approach, evolving environments specifically optimized for ideal agent learning. Though these works evolve the environment rather than the agent, they provide inspiration for our work as they highlight how environmental complexity can elicit sophisticated behaviors.

Most closely related to our work are several efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task where a small number of hider agents learn to use environmental tools to evade seeker agents, both trained via reinforcement learning. Other works similarly train populations for specific tasks: playing video games (Jaderberg et al. 2019), manipulating objects (Petrenko et al. 2021), or capturing agents (Zheng et al. 2017; Yamada et al. 2020). However, while many of these works explore emergence, they do so using fixed tasks and objectives. The Neural MMO framework (Suarez et al. 2019 2023) takes a more open-ended approach to reinforcement learning, placing agents within a massively multi-agent game setting and optimizing only for survival rather than fixed objectives. However, the population remains capped at a fixed number of agents (experiments examine up to 128 concurrent agents) without generational inheritance, limiting examination of population dynamics; by contrast, our populations exceed 60,000 concurrent agents, more than two orders of magnitude larger, and

are free to grow and shrink across generations. [Hamon et al. \(2023\)](#) aligns more closely with our work through variable population sizes, genetically inspired evolution, and generational timelines, though its environment is significantly smaller, less feature-rich and it does not directly explore the importance of scale, precisely the variable we isolate and measure. Similarly, [Lu et al. \(2024\)](#) also simulates agents evolving to gather resources. However, this work too operates at a far smaller scale (a maximum of 256 agents, roughly $250\times$ fewer than ours) and emphasizes tool use through programmable robots. While all of these past works take steps towards examining emergent ecological dynamics, comprehensive understanding requires populations that can grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can emerge. Our work integrates these elements at a scale that lets us measure, rather than merely observe, how emergent behavior depends on population and environment size, enabling examination of ecological phenomena inaccessible in prior work.

Finally, we are also inspired by recent works that emphasize environmental speed and scale, using optimized libraries to increase the computational speed and number of agents able to interact in game-like environments ([Koyamada et al. \(2023\)](#), [Matthews et al. \(2024\)](#), [Lechner et al. \(2023\)](#)). We build upon this progress in computational efficiency, utilizing similar JAX-based acceleration techniques to achieve large scales and fast simulation in ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We model the interactions between populations of agents and their environment as a type of multiplayer game we refer to as an **Ecological Game (EG)**. Similar to mathematical models such as Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see [Sutton et al. \(1998\)](#) and [Albrecht et al. \(2024\)](#) for reference), an EG contains underlying Markovian transition dynamics but is designed to express changing populations of agents in an objective-free setting. Like an SG, an EG contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and a set of actions for each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Unlike an SG, it removes the reward function, and adds a population function $\Gamma(s_t)$ that provides information about which agents in I are currently alive and the identity of their parents so that birth and death events can be inferred. Just as MDPs and SGs have partially observable counterparts, POMDPs and POSGs, an EG can also be partially observable, in which case we refer to it as a POEG. The difference between a POEG and an EG is that a POEG does not allow each agent to observe the entire state, and instead incorporates an observation function $O(s_t, i)$ that maps a state and player to an observation for that player. Our environment is a POEG with different observation functions depending on various sensor configurations explained in the next subsection.

3.2 ENVIRONMENT

The particular POEG used in our experiments is a large grid world containing various resources that agents must collect in order to reproduce and survive. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ where s_t^m is 2D data associated with the map, such as the terrain and distribution of resources, and s_t^a is data associated with the agent, such as its position, health, and internal resources.

Terrain: Each map has a rock layer defining the height of each grid cell. Maps also include water, which is initially added to grid cells with a rock value below a preset sea-level. Throughout an episode, water flows downhill using a discrete flow simulation based on the height defined by this rock layer. In the experiments below, we use six global terrain configurations shown in Figure 2: **Ocean, Beach, Island, Lake, Isthmus, and Channel**. We sweep five grid sizes— 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 —so that each step up the ladder quadruples the number of grid cells, giving us a controlled $256\times$ range in area over which to measure scaling effects. When growing or shrinking the grid size, we scale the initial population size proportionally to the change in grid area, keeping the initial density of agents fixed so that only scale, not crowding, varies across

conditions. Once the simulation begins running, the population then fluctuates according to the dynamics of the environment.

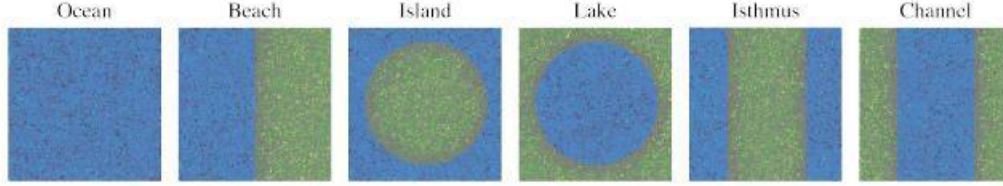


Figure 2: Overhead images of the terrains used in our experiments at 256×256 grid size. Small dark spots represent individual agents, while green and yellow represent concentrations of energy and biomass. These images were each captured 5000 steps into an episode after most biomass in the water has been eaten, but before the agents have evolved to survive on land for longer periods.

Resources and Agent Health: Agents must collect **water**, **energy**, and **biomass** to survive and reproduce in these environments. Initially, biomass and energy are distributed randomly throughout the environment but can be consumed and redistributed by individual agents.

Each action an agent takes expends a small amount of energy and water. The spent water returns to the landscape, where it flows downhill to the nearest body of water, while the energy is used up and destroyed. Biomass is used for reproduction; all agents require a fixed amount of biomass to exist and therefore cannot reproduce until they have collected enough excess biomass to donate to their offspring. Like water, the total biomass remains constant in the environment throughout the simulation, but unlike water, the biomass does not flow downhill or move on its own. Energy is not held constant but slowly grows on free-standing biomass. We designed these resource dynamics to provide agents with a slightly more challenging problem than collecting a single food source. For ablations on resource rules, see Appendix F.2

Agents also have health points (HP) that they must maintain in order to avoid death. If an agent overspends a critical resource, their health will rapidly deteriorate, but they may use energy and water to slowly recover HP over time. Agents may also attack other agents, reducing the targets' HP. They also take a small but increasing amount of damage based on their age, meaning that agents eventually die of old age if not by other causes. When an agent's HP goes to zero and they die, whatever resources they were carrying are returned to the environment in the grid cell they were standing on. In this way, if an agent attacks and kills another agent, it may then consume their resources as a basic form of predation. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents sense the environment using four sensors. The **internal** sensor tells the agent about its age, health, and the resources it is carrying; the **external** sensor detects external resources in the same grid cell as the agent; a **compass**, inspired by the magnetoreceptors found in birds and other animals, detects the agent's global direction as a 4-way one-hot vector; and **vision** provides a 7×7 top-down rendered patch around the agent, containing three color channels as well as a fourth channel showing the local difference in elevation. For ablations on vision configurations, see Appendix F.6

Actions: Agents have a finite set of discrete actions grouped into five categories. The first is **Rest**, which does nothing but costs the least energy. There are three **Move** actions: rotate clockwise, rotate counterclockwise, and move forward one grid cell. Multiple agents cannot occupy the same cell; if an agent tries to move into the same cell as another agent, or if two agents try to move to the same cell, their move action is canceled. The agents also have a single **Attack** action which kills all agents in the 3×3 square directly in front of them. For ablations on attack strength, see Appendix F.7. There are three separate **Eat** actions, one for each of the resource types. If an agent is in the same cell as a particular resource, taking the corresponding eat action will transfer a fixed amount of that resource from the environment to the agent's stomach, which has a fixed maximum capacity.

Finally, agents have a single **Reproduce** action that creates a new offspring via single-parent reproduction. This action is only successful if the agent has accumulated enough resources to donate to its new child. The newly created agent is created directly behind its parent. The child's policy contains a mutated copy of its parent's weights. The child also immediately inherits a fixed amount of its parent's resources.

In the experiments below, our central comparisons are between agent classes that are identical except for a single added ability, which lets us attribute any difference in emergent behavior to that ability alone. We use three agent sensing combinations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). We use **+A** to note when attacking is allowed; for example (**RC+A**) indicates the agent has resource sensors, a compass, and can attack. Thus the (**R**) vs. (**RC**) contrast isolates the effect of the compass, and the (**RC+A**) vs. (**RCV+A**) contrast isolates the effect of vision.

3.3 POLICIES

Our agent policies are implemented as small, memoryless MLPs. We opted for agents without memory in order to simplify the computational model, and reduce network parameters. The raw values for each sensor are flattened, normalized to the range $[-1, 1]$ and fed through separate linear layers with an output dimension of 64. These values are summed and fed through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. The action space is discrete, so the final output is a linear layer with an output dimension equal to the number of possible actions. Actions are sampled from a softmax distribution with a temperature that is allowed to evolve with the network weights. Weights are stored in half-precision bfloat16, which is what lets us hold populations of tens of thousands of policies in memory at once while keeping run times low. Each agent’s network is deliberately small—roughly 10k–25k parameters depending on active sensors, i.e. orders of magnitude below a typical deep RL policy—so that scale can be spent on population and world size rather than per-agent capacity. Appendix F.5 shows this behavior is stable across a $\sim 8k$ –78k parameter range.

In all runs, the weights for the initial population of agents use Kaiming initialization (He et al., 2015); bias vectors are initialized to zero. Policies for agents born in the middle of an episode are created by copying their parent’s weights and applying Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation approach is similar to that used in other neuroevolution research (Such et al., 2017). While more complex approaches such as crossover are possible, we opt for this mechanism to simplify agent reproduction and study the effect of environmental scale in as simple a setting as possible. Empirically, we found that a larger mutation rate was necessary due to the use of low-precision floating-point numbers. We perform an ablation on mutation rates in Appendix F.4.

Agents also have a 3-channel trait controlling their color when rendered onto the map. The color trait mutates similarly to the network weights and determines what an agent looks like when observed by other agents with the vision sensor. In theory, this allows agents to distinguish between different types of other agents.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and run on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Every experiment runs for a full 2M environment steps or until extinction unless otherwise noted, and each configuration is repeated across four random seeds (with 32 or 64 seeds in our survival analysis) so that the comparisons we report reflect consistent patterns rather than single runs. On a single H100 GPU, per-seed wall-clock times range from under an hour for 64×64 grids to approximately 10 hours for the largest 1024×1024 grids, keeping even our largest-scale sweeps within a practical hardware budget. Appendix E describes simulator architecture in more detail, reports additional scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

In the experiments below, we quantify the effects of three variables—sensing modality, environmental scale, and population size—on the emergent behavior of agents, in each case comparing conditions that differ only in the variable under study.

4.1 LONG-DISTANCE RESOURCE GATHERING

In our first experiment, we measure the emergent ability of agents to travel long distances inland in order to find resources. The comparison is deliberately clean: in environments with clear directional structure, agents equipped with a compass (**RC**) develop far more effective long-range resource gathering than an otherwise-identical baseline class (**R**) without a compass, and the gap is both large and highly consistent across seeds. We further show that the emergence of this behavior grows monotonically more reliable and more pronounced as environmental scale increases.

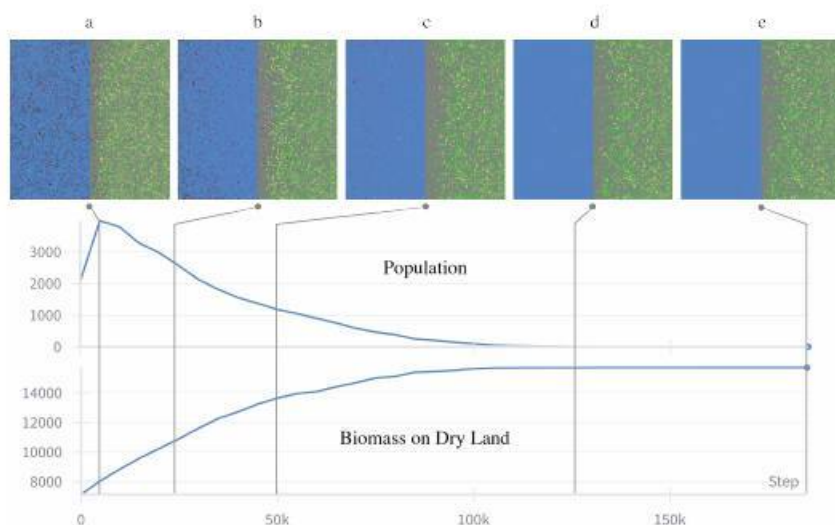


Figure 3: A representative trial with baseline (**R**) agents (no compass) in a 256×256 **Beach** environment. The top chart tracks population over time; the bottom chart tracks the free biomass on dry land. Without a compass, the population collapses to extinction as biomass is irreversibly transferred to land—the failure mode that the compass condition in Figure 4 avoids.

We begin with a visual example of the baseline (**R**) agent class, shown in Figure 3 which establishes the failure mode against which the compass condition is measured. After 5000 time steps (a), all the initial agents on dry land have died due to lack of water, while those in the water have rapidly increased in number due to the rich initial resources. At this early stage, the population does not yet contain sophisticated policies that can efficiently seek out resources. From this early peak, agents in the water occasionally stumble onto land (b). When they do, they typically cannot find their way back to the ocean and die, depositing their biomass on land. By 50,000 time steps (c), this gradual process has nearly doubled the uneaten biomass on land. This one-way transfer of resources from water to land shrinks the population, because the water can no longer supply enough biomass to sustain it. By 125,000 steps (d), almost all biomass has been transferred to land and the population is nearly extinct, with the final agent dying around 185,000 steps (e)—so this baseline run ends in total extinction well before 10% of the 2M-step budget elapses.

This example demonstrates the ecological impact of the agents’ inability to find their way back to water. Not only do the individuals die, but their deaths alter the distribution of resources in the environment in ways that have important consequences for the entire population. At the same time, this ecological catastrophe presents an opportunity: any lineage that discovers how to walk onto land, forage in the rich deposits of biomass left by its dead ancestors, and safely return to water can convert this failure mode into a stable resource supply. The remainder of this section quantifies exactly how reliably each agent class seizes that opportunity.

In the matched comparison run of Figure 4 the only change is that agents are given an internal compass (**RC**). Early on, the two conditions look alike (a): the population shrinks and free biomass on land rapidly accumulates. From this shared starting point the trajectories diverge sharply. Some compass agents drift into the relative safety of the corners, and by around 150,000 steps (b) they begin to “mine” the biomass on land, using the compass to shuttle back and forth between land and water. By 200,000 steps (c) these miners have cleared the entire beach. The population then only

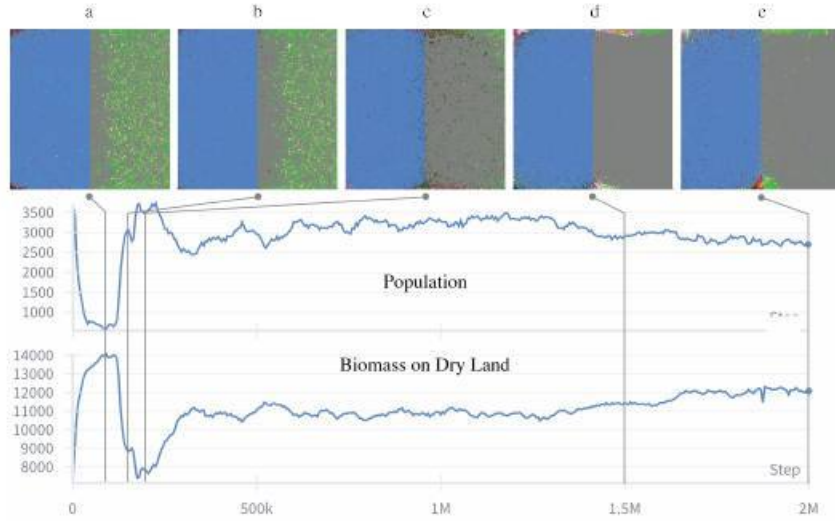


Figure 4: A representative trial with compass-equipped (**RC**) agents in a 256×256 **Beach** environment—the same setup as Figure 3 with only the compass added. The top chart tracks population over time; the bottom chart tracks the free biomass on dry land. In contrast to the baseline, mining clears the accumulated land biomass and the population persists stably for the entire 2M-step run.

fluctuates mildly, forming clumps around the edges (d), and persists for the full 2,000,000 time steps (e). The contrast is stark: where the baseline in Figure 3 went extinct before 185,000 steps, the single added compass sensor yields a population that survives an order of magnitude longer, to the end of the run.

Table 1: The compass advantage on the **Beach**, and its growth with scale (four seeds per cell). We compare agents that differ only in the compass sensor—(**RC**) vs. (**R**)—across five world sizes and matched initial populations. counts seeds that developed mining; counts seeds that went extinct before 2M steps. Reading across, (**RC**) agents mine in all four seeds at every size from 256×256 up, versus at most 1/4 for (**R**) at any size, and (**RC**) extinctions vanish (0/4) at the three largest scales while (**R**) populations collapse in all four seeds up through 512×512 . See Table 3 in Appendix A for per-seed detail.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

We quantify these observations across the full five-point scale sweep, running **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 with initial populations of 32768, 8192, 2048, 512, and 128 agents respectively. Each configuration is run with four random seeds; we score a mining event as a drop of at least 10% in the free biomass on dry land from its historical high point, and also record the onset time of the event (the local maximum in land biomass) and the size of the biomass drop. Table 1 summarizes mining and extinction outcomes, with per-seed detail in Table 3 in Appendix A. The comparison is decisive. Compass (**RC**) agents evolve mining in 17 of 20 seeds overall, whereas the matched (**R**) baseline does so in only 1 of 20—and that lone case appears exclusively at the largest scale. Reliability rises monotonically with scale for (**RC**) agents: mining occurs in all four seeds at every size from 256×256 up to 1024×1024 , drops to 3/4 at 128×128 , and to 2/4 at the smallest 64×64 world. Extinction shows the mirror-image trend: (**RC**) populations never go extinct at or above 256×256 (0/4 at each of the three largest sizes), but

fail in 1/4 and 3/4 of seeds at 128×128 and 64×64 . The baseline is far more fragile still, going extinct in all four seeds at every size from 64×64 through 512×512 .

To put these effects on a firmer statistical footing, we run a larger survival study—32 seeds each of **(R)** and **(RC)** agents at 512×512 and 64 seeds each at 128×128 , for 192 runs in total—and use a Kaplan-Meier estimator to measure the probability of extinction and of mining as a function of simulator time. At the larger world size, extinction is both rarer and slower to occur, while in the raw Kaplan-Meier curves mining is more frequent and happens sooner at the smaller grid size. A Cox proportional hazards model, which adjusts for both covariates jointly, quantifies the trends with tight confidence intervals (all $p < 0.005$): moving from 128×128 to 512×512 reduces the extinction hazard about 25-fold ($HR=0.04$) and raises the mining-discovery hazard roughly 6-fold ($HR=6.07$), while enabling the compass reduces the extinction hazard by a comparable amount ($HR=0.03$) and accelerates mining emergence by roughly 35-fold ($HR=34.59$). The model separates the groups well (concordance 0.80 for extinction, 0.85 for mining). Full details are in Appendix B.

Table 2: The compass advantage generalizes across terrains (four seeds per cell, 512×512). Comparing **(RC)** against otherwise-identical **(R)** agents in the **Island**, **Lake**, **Isthmus**, and **Channel**, the compass yields at least as many mining seeds (🔍) and no more extinctions (💀) in every terrain. The gap is widest on the **Channel** (3/4 vs. 0/4 mining; 1/4 vs. 4/4 extinct) and **Isthmus** (4/4 vs. 1/4 mining; 0/4 vs. 3/4 extinct). 💀 counts seeds extinct before 2M steps. See Table 4 in Appendix A for per-seed detail.

	Island		Lake		Isthmus		Channel	
	🔍	💀	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

To test whether the compass advantage generalizes beyond the **Beach**, we repeat the **(R)** vs. **(RC)** comparison at the 512×512 grid size across four additional terrains: **Island**, **Lake**, **Isthmus**, and **Channel**. Mining and extinction outcomes are summarized in Table 2, with per-run detail in Table 4 in Appendix A. The compass advantage holds in every terrain: in each of the four, **(RC)** agents mine at least as often as—and typically far more often than—their **(R)** counterparts, discover mining earlier on average, and go extinct less frequently. The margin is largest on the **Channel**, where **(RC)** agents mine in 3/4 seeds while the baseline mines in 0/4 and goes extinct in all four, and on the **Isthmus** (4/4 vs. 1/4 mining; 0/4 vs. 3/4 extinction). **(RC)** agents adapt most easily to the island and isthmus (4/4 mining, 0/4 extinction in both), and struggle only on the lake and channel, where they occasionally go extinct. We hypothesize that walking straight east or west always returns an agent to water on the island and isthmus, whereas the channel and lake admit no such simple rule. The **(R)** baseline never adapts reliably to any of these terrains, faring best on the island (3/4 mining) and worst on the channel (0/4 mining, 4/4 extinction). See Figure 12 in Appendix D for a visual of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We next turn to the **Ocean** environment to measure whether agents adapt to use vision to locate free resources and prey on other agents. As in the compass study, the comparison isolates a single sensor: **(RCV+A)** agents with resource sensing, a compass, and a 7×7 visual sensor against otherwise-identical **(RC+A)** agents that lack only vision.

Figure 5 plots population, biomass utilization, movement actions, and eat actions for both agent classes at the 512×512 and 128×128 grid sizes. Plots for the other grid sizes appear in Figure 9 in Appendix C, with per-seed summary statistics in Table 5 in Appendix A. Biomass utilization is the percentage of the environment’s total biomass that is currently held inside agents. Across every scale, vision (red lines) produces a consistent behavioral signature: agents move less and eat more, and the effect is large. Averaged over seeds at 512×512 , vision agents reach a mean population of 16,345 versus 7,323 for blind agents (a $2.2\times$ increase) and a biomass utilization of 0.368 versus 0.189 (nearly double), while spending a smaller fraction of actions moving (0.256 vs. 0.416) and

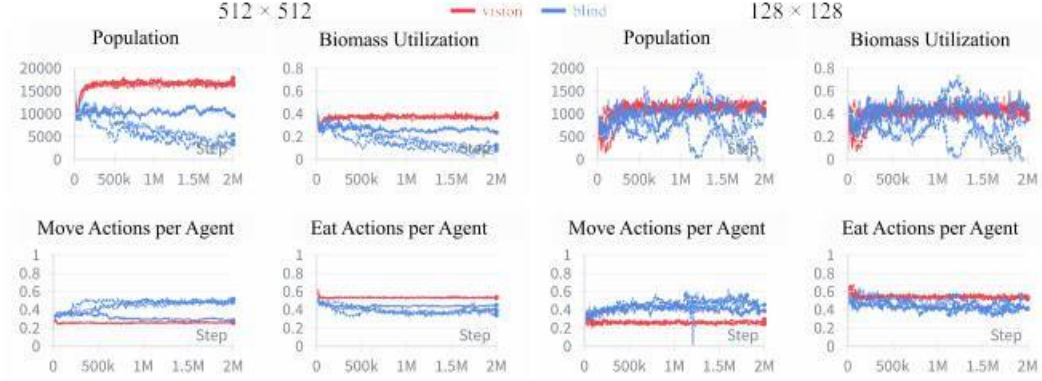


Figure 5: Foraging in 512×512 and 128×128 **Ocean** worlds: population, biomass utilization, move actions per agent, and eat actions per agent, each averaged across agents at every time step. Each panel overlays four vision seeds (**RCV + A**) (red) against four blind seeds (**RC + A**) (blue), holding all else equal. The vision curves sit consistently above the blind curves in population and biomass utilization and below them in movement, reflecting agents that move less and eat more effectively; at 512×512 this amounts to roughly double the population and biomass utilization of the blind baseline (Table 5). The vision advantage also grows cleaner at larger scale, with markedly less variance across seeds—evidence that larger environments make vision an increasingly decisive advantage over blindness for foraging. See Figure 9 in Appendix C for grid sizes 1024×1024 and 256×256 .

a larger fraction eating (0.532 vs. 0.402). The same ordering holds at all four scales in Table 5 indicating that agents reliably exploit vision to forage more efficiently. The advantage also becomes cleaner with scale: variance across seeds shrinks substantially in larger worlds, and, notably, the vision runs are the more stable of the two conditions, varying less both across seeds and over time.

Figure 6 reports averaged attacks per agent and homicides per attack for the same runs, revealing that vision drives a distinct predation strategy: attack rarely, but kill efficiently. Vision agents take the attack action infrequently yet convert those attacks into kills at a remarkably consistent rate—roughly 0.73–0.78 homicides per attack across all four scales, i.e. a kill rate near 75%. Blind agents attack several times more often (for example, 0.045 vs. 0.015 attacks per agent at 512×512) but land those attacks far less reliably, with per-scale mean kill rates ranging from 0.24 to 0.85 that are also much more volatile from seed to seed; in large grids they occasionally spike to high kill rates but with highly unstable dynamics. Confounding factors make the precise cause hard to pin down. Visual inspection (Figure 13 in Appendix D) shows that such spikes tend to arise when agents pile into clusters in one location—clusters that almost never form under vision, suggesting that vision agents also adapt to avoid one another. In smaller environments the vision–blind gap in these predation metrics narrows, but vision populations remain far more stable throughout.

4.3 ABLATIONS

To establish how robust these quantitative trends are, Appendix F sweeps the key simulator settings one at a time—initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8)—and reports how mining, extinction, and foraging metrics respond to each. The headline scaling effect survives these controls, and Appendix E.3 further confirms, by matching total cells and agents across conditions, that the stability gains at large scale cannot be explained away as mere self-averaging over many small worlds.

5 CONCLUSION

Our study highlights the promise of open-ended ecological settings as a powerful paradigm for investigating the emergence of intelligence. Through controlled experiments that vary one factor at a time across a $256 \times$ range of world areas, we find that environmental scale and sensory richness

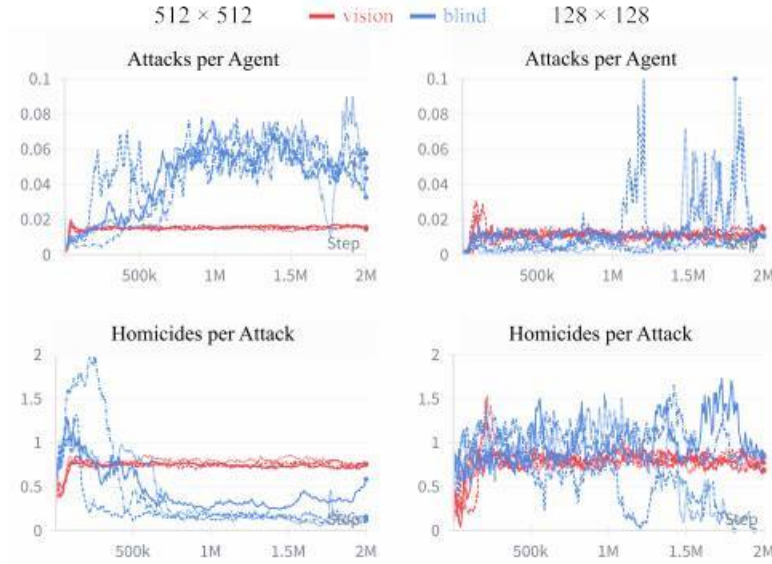


Figure 6: Predation in 512×512 and 128×128 **Ocean** worlds: attacks per agent and homicides per attack, averaged across agents. Each panel overlays four vision seeds (**RCV + A**) (red) against four blind seeds (**RC + A**) (blue). Vision cleanly separates the two: vision agents attack several times less often but convert attacks into kills at a near-constant $\sim 75\%$ rate, whereas blind agents attack more but kill less reliably and far more erratically—the signature of specialized, precise predation. The separation sharpens at larger scale. See Figure 10 in Appendix C for grid sizes 1024×1024 and 256×256 .

both contribute, by large and consistent margins, to the development of sophisticated sensing and decision-making policies. The single-sensor comparisons make this concrete: adding a compass turns near-universal extinction into reliable, long-lived mining (mining in 17/20 vs. 1/20 seeds on the **Beach**), and adding vision roughly doubles sustained population and biomass utilization while yielding a precise $\sim 75\%$ predation kill rate—with the reliability of both effects improving monotonically as the world grows, and a survival analysis over 192 runs attaching hazard ratios of up to $\sim 35\times$ to these factors. Taken together, these results suggest that ecological and evolutionary pressures, when coupled with rich environments, may provide a natural pathway for the development of increasingly intelligent artificial agents. Furthermore, our experiments demonstrate that these rich behaviors can be evolved in populations of tens of thousands of agents within a reasonable hardware budget—on the order of ten H100-hours per large run—and without any explicit objectives. We hope that these results will enable further exploration of ecology as a mechanism for developing and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Per-seed mining and extinction outcomes on the **Beach**, comparing compass (**RC**) against otherwise-identical compass-free (**R**) agents across five scales (four seeds each). The per-cell averages at the foot of each block make the compass advantage and its scaling explicit.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x			140	✓			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			
2	x			n/a	x			1070k	x			165k	x			55k	x			
3	x			n/a	x			280k	x			125k	x			90k	x			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			

Table 4: Per-seed mining and extinction outcomes across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains, comparing compass (**RC**) against otherwise-identical (**R**) agents (four seeds each). Grid size is 512×512 with an initial population of 8192. Across all four terrains, (**RC**) agents mine at least as often, discover mining sooner on average () , and go extinct no more often than (**R**).

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging compared for Vision (**RCV+A**) and Blind (**RC+A**) agents in the **Ocean** across four scales (four seeds each).  is population size;  is biomass utilization (percentage of the system's total biomass currently inside agents' stomachs);  is move actions per agent (fraction of time spent moving, averaged across agents);  is the corresponding proportion of eat actions; all are averaged over all steps for the seed. At every scale, vision agents post higher  and  and higher  with lower , reflecting agents that move less and eat more; at 512×512 the per-seed averages show roughly double the population (16,345 vs. 7,323) and biomass utilization (.368 vs. .189) of the blind baseline.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation compared for Vision (**RCV+A**) and Blind (**RC+A**) agents in the **Ocean** across four scales (four seeds each). 🗡️ is attacks per agent (fraction of time an agent attacked, averaged across agents); 🎯 is homicides per attack (agents killed per attack action); both are averaged over all steps for the seed. The contrast is consistent: vision agents attack several times less often (e.g. 🗡️ .015 vs. 🗡️ .045 at 512×512) yet kill far more reliably, holding a near-constant $\sim 0.73\text{--}0.78$ homicides-per-attack across all scales, whereas blind kill rates ($\sim 0.24\text{--}0.85$) swing widely from seed to seed. Note the outlier of 0.562 for 🗡️ of seed 3 in the 128×128 (**RC+A**) run: this seed went extinct before 2M steps, so its shrinking population inflates the attacks-per-agent metric.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

To place the compass and scale effects on a rigorous statistical footing, we compare rates of extinction and mining emergence as a function of two crossed factors: world size and access to global direction via the compass sensor. Extinction is defined as the population reaching zero individuals, and mining is defined exactly as in the main experiments (a 10% drop in free biomass on land).

This yields a 2×2 design spanning 192 runs: 64 runs each of 128×128 with and without compass, and 32 runs each of 512×512 with and without compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution (treating runs surviving past 2M steps as right-censored). Figures 7 and 8 show a clear and consistent ordering: whether we enlarge the world or add the compass, populations survive longer and discover mining sooner, and combining both factors is best of all.

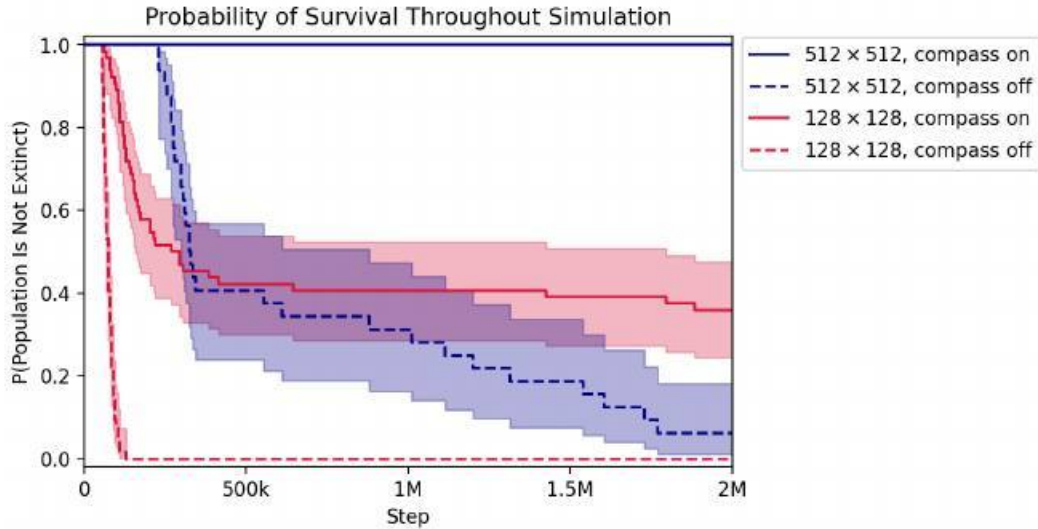


Figure 7: Kaplan-Meier survival curves: probability the population has not gone extinct as a function of timestep, for each combination of world size and compass access. A separate estimator is fit per group (32 runs per 512×512 group, 64 runs per 128×128 group), with 95% confidence intervals shaded. The curves separate cleanly and monotonically—both a larger world and an added compass shift survival upward—and the two 512×512 curves dominate the 128×128 curves throughout.

We quantify the timing of emergence and extinction with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7; runs without an event by 2M steps are censored). Every effect is large and highly significant ($p < 0.005$, tight confidence intervals). For extinction, both larger world size and compass use sharply reduce the hazard: enlarging the world from 128×128 to 512×512 makes extinction about $25\times$ rarer ($HR=0.04$), and enabling the compass produces a comparable $\sim 33\times$ reduction ($HR=0.03$). For mining emergence the same two factors act in the accelerating direction: larger worlds discover mining about $6\times$ faster ($HR=6.07$), and the compass accelerates discovery even more dramatically, by roughly $35\times$ ($HR=34.59$). The models discriminate between groups well (concordance 0.80 for extinction, 0.85 for mining). We attribute these effects to larger worlds sustaining larger populations, which both lowers the chance of hitting the absorbing zero boundary and, through a greater total volume of mutation, explores a larger phenotype space than smaller worlds.

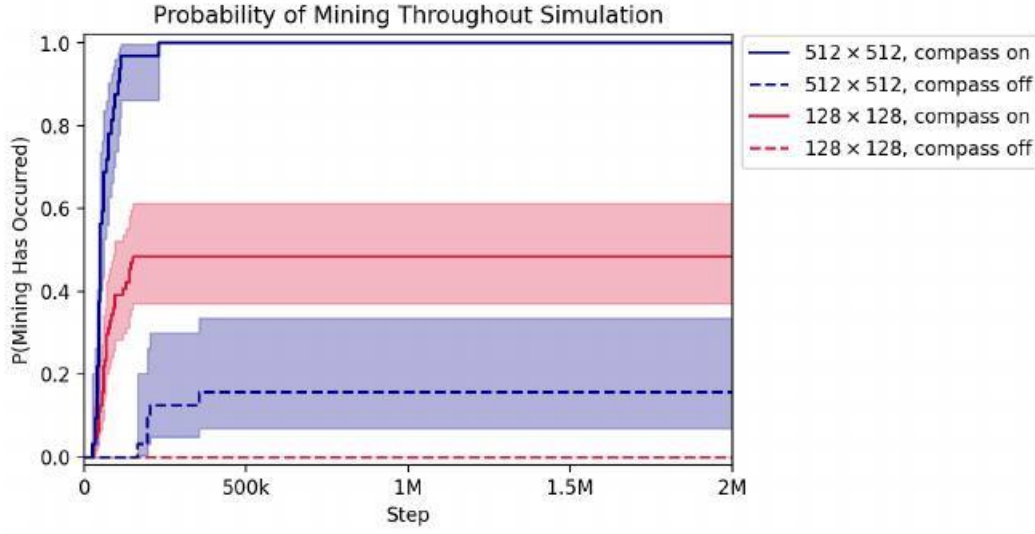


Figure 8: Kaplan-Meier emergence curves: probability that mining has evolved within the population as a function of timestep, for each combination of world size and compass access. A separate estimator is fit per group (32 runs per 512×512 group, 64 runs per 128×128 group), with 95% confidence intervals shaded. Adding the compass moves the curves sharply earlier and higher at both scales, the single largest driver of faster mining discovery in this experiment.

Table 7: Cox proportional hazards models for extinction and mining, relative to a 128×128 , compass-off baseline (192 runs: 32 compass-on and 32 compass-off seeds at 512×512 ; 64 and 64 at 128×128). Hazard ratios (HR) below 1 mean reduced hazard, i.e. longer time to event: extinction $HR = 0.04$ for 512×512 means extinction occurs at only $0.04 \times$ the 128×128 rate (a $\sim 25 \times$ reduction), while a mining $HR = 34.59$ for compass means mining is discovered at roughly $35 \times$ the baseline rate. Every covariate is significant at $p < 0.005$, and both models show good discrimination (concordance 0.80 and 0.85).

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

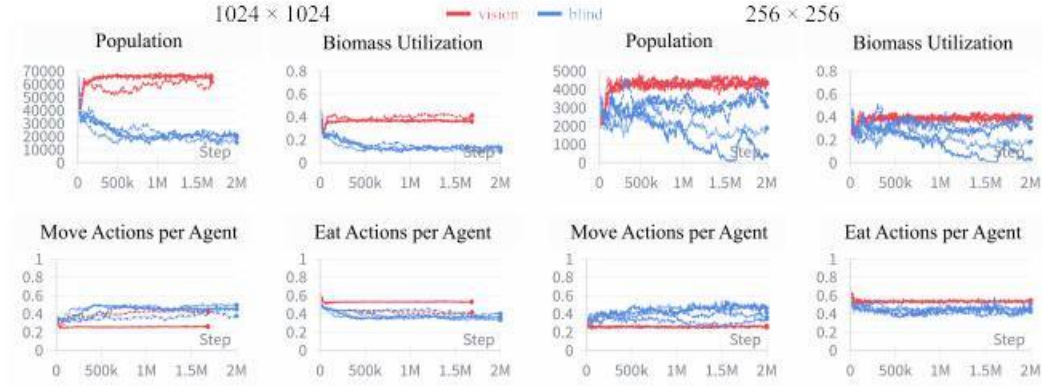


Figure 9: Foraging in the **Ocean** at the 1024×1024 and 256×256 scales. The same ordering holds across world sizes: vision agents (red) sustain larger, more stable populations and higher biomass utilization by eating more and moving less, and the vision advantage grows cleaner at the larger scale with less variance across seeds. Due to GPU-hour limits, the vision seeds at 1024×1024 ran for 1.7M rather than 2M steps.

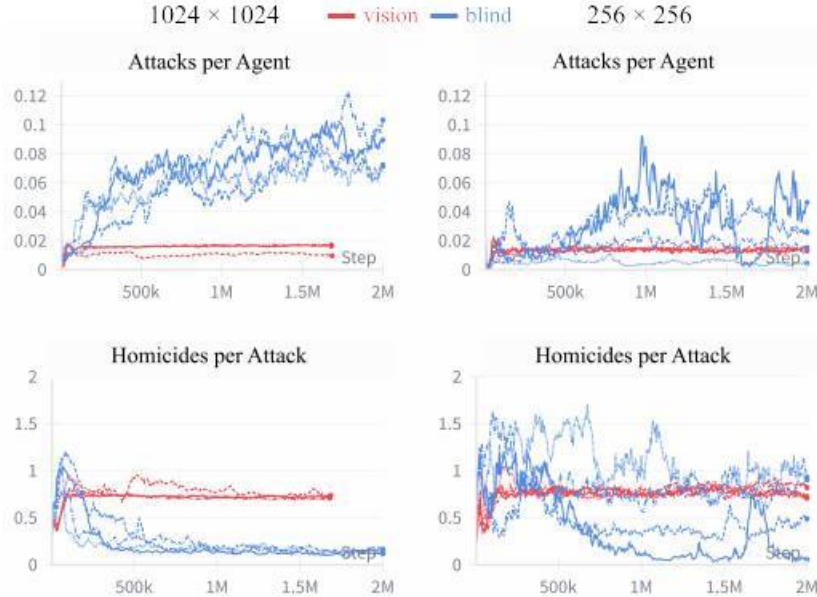


Figure 10: Predation in the **Ocean** at the 1024×1024 and 256×256 scales. As at the other scales, vision agents (red) attack much less often but land kills with far higher and steadier precision, the signature of specialized predation layered on top of foraging (Figure 9). The separation is sharper at the larger scale, while blind agents (blue) grow markedly more erratic in the smaller world.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

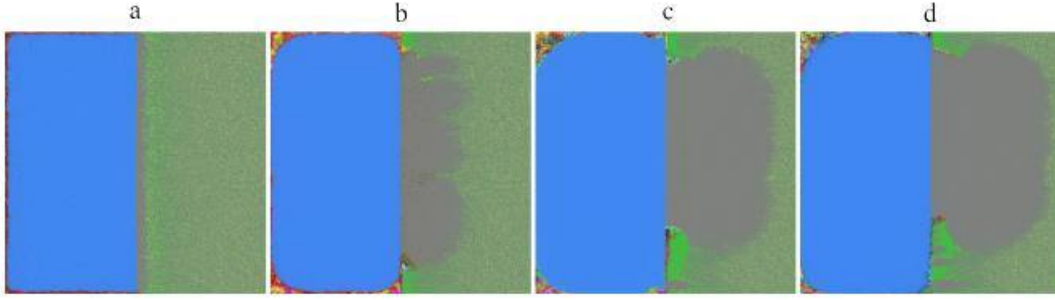


Figure 11: At the largest 1024×1024 scale, richer ecologies give rise to visual patterns not seen in smaller worlds. Here **(RC)** agents first died at the shore (a), transferring biomass to dry land, then evolved mining and cleared the beach (b). When biomass re-accumulated at the ends of the beach (c), subpopulations independently re-evolved mining, traveling inland along these secondary biomass beaches to extract the resources (d)—an example of the repeated, self-sustaining mining that emerges most reliably at this largest scale.

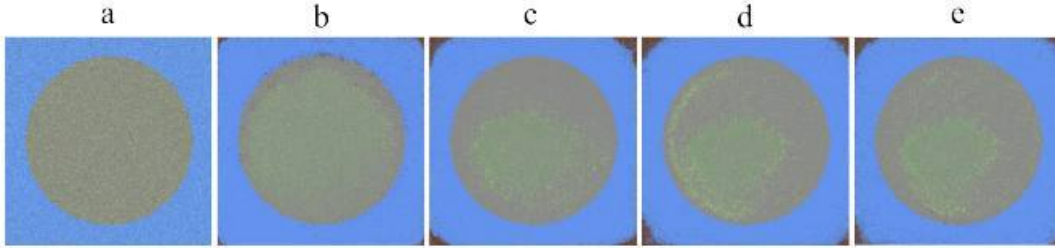


Figure 12: Mining generalizes beyond the **Beach**: a 512×512 **Island** run with **(RC)** agents, the terrain where the compass advantage is largest (4/4 mining, 0/4 extinction). Different regions of the shore are mined over the course of the run (b, c). Biomass is transferred back onto the island once mining becomes difficult and agents die before returning to water (d), and agents then adapt to re-mine this wall of accumulated biomass, clearing it from shore (e).

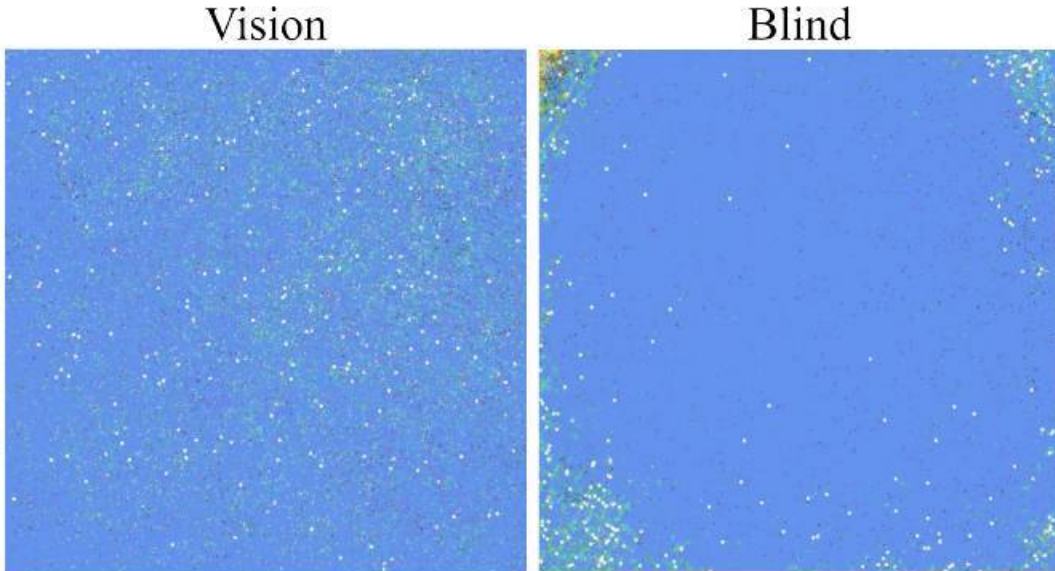


Figure 13: Side-by-side comparison of a 512×512 **Ocean** run with attacking enabled: vision **(RCV+A)** agents on the left, blind **(RC+A)** agents on the right, with white squares marking attack locations. The visual difference tracks the quantitative one: vision agents spread out to locate resources across the whole ocean and prey more effectively (see Figures 5 and 6), while blind agents explore far less and instead cluster in the corners—the very clustering that inflates the sporadic high blind kill rates noted in the main text.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report details on simulation design, runtime, and experiment reproducibility.

Simulator design. We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation allows the same environment implementation to be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

A key feature of the simulator is its built-in tooling, including a measurement API for computing and recording aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a concrete gap relative to existing systems. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), or Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. Our platform instead combines GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, and this combination is what enables the open-ended populations of over 10^5 agents used throughout this paper while remaining interoperable with diverse population algorithms. The result is a state-of-the-art platform for scaling ecological experiments like ours and a foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 quantifies throughput and wall-clock cost across the full grid-size sweep on a single H100 GPU (2M steps max per run), for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total); runtime varies with network size, active sensors, and logging level. Throughput scales smoothly with world size: the **Beach** simulator sustains 750 steps/s at 64×64 and still 55 steps/s at 1024×1024 , a $256 \times$ larger world running only $14 \times$ slower per step.

Even our largest runs remain affordable: 1024×1024 completes in ~ 10 hours on the **Beach** and ~ 14 hours in the **Ocean** on one H100, at roughly 60GB of memory. **Ocean** runs are modestly slower (e.g. 40 vs. 55 steps/s at 1024×1024) owing to the enabled attack action and vision sensing.

Reproducibility. Because our environments are large and randomized, the precise trajectory of any single run varies substantially from run to run, which is exactly why every comparison in this paper is reported over at least four random seeds (and 32–64 seeds in the survival analysis). Repeating an experiment even with an identical random seed does not reliably reproduce the same trajectory: modern GPUs may execute parallel floating-point operations such as summations in different orders and fuse operations in ways that alter rounding, and these tiny finite-precision differences accumulate over the millions of steps in a run into noticeable drift. The 16-bit weights coarsen but do not dominate this effect. Crucially, this run-to-run variation is confined to trajectory detail: when seeds are aggregated, the high-level comparisons we report—the compass and vision advantages and their growth with scale—emerge as clear, consistent patterns rather than seed-specific artifacts.

Table 8: Throughput (simulation steps per second, hz) and extrapolated wall-clock runtime (hours for 2M steps) in the **Beach** and **Ocean** worlds across the full grid-size sweep, on a single H100. (RC) agents use a 22,465-parameter network and (RCV + A) agents a 22,529-parameter network. Throughput degrades gracefully with scale—a $256\times$ increase in world area costs only about a $14\times$ slowdown per step—keeping even the largest runs to ~ 10 – 14 hours. Seeds that go extinct before 2M steps finish sooner still.

	1024×1024		512×512		256×256		128×128		64×64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

F ABLATIONS

We present a series of controlled single-variable ablations that both quantify how each hyperparameter affects the emergent-behavior metrics and test how robust the main-text findings are to changes in environment and agent configuration. Two conclusions recur throughout: the headline scaling and sensor effects survive these sweeps, and where a setting does matter its effect is smooth and interpretable rather than knife-edge.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We isolate the effect of initial population size, holding world size fixed at 256×256 **Beach**, to test whether the scaling trends of the main text are driven by population count rather than physical scale. Results are for compass (RC) agents; the (R) baseline again performs near the floor, with 0% mining and 100% extinction at every initial population size except the largest (8192), where it reaches only 25% mining and still 75% extinction. (Mining and extinction frequencies need not sum to 100%: a population may mine and then go extinct, or survive all 2M steps without ever mining.)

Figure 14 shows that, in sharp contrast to physical scale, initial population size has essentially no effect: populations seeded with just 128 agents evolve mining as often as those seeded with 8192, a $64\times$ larger start, and extinction rates are flat across the range. This cleanly separates the two variables—it is world size, not the starting head-count, that drives the trends in Table 1.

Figure 15 explains why the effect is so small: regardless of starting size, every run converges to a similar population within the first 50,000 steps. The environment thus imposes an implicit carrying capacity set by its fixed resources—populations below it grow toward it, and populations above it are pulled down to it—so that after this early correction the effective population size is nearly identical across conditions, leaving mining and extinction frequencies unaffected. The one residual trace of initial size is on the maximum population reached: because agents carry biomass that is deposited on death, larger starting populations add slightly to total resources and reach modestly higher peak populations (Table 9), but the effect is far too small to move mining or extinction frequency in this setting. Note also that convergence early on does not imply constancy: later in a run, populations across seeds can again diverge substantially as agents evolving different abilities at different times reshape the resource distribution.

F.2 VARYING RESOURCE RULES

We quantify how two resource hyperparameters—biomass photosynthesis rate and initial biomass site density—affect mining and extinction frequency, using the same 256×256 **Beach** setup as Appendix F.1 with initial population fixed at 2048 and compass (RC) agents. Sweeping these against the null result for initial population size lets us cleanly separate the knobs that govern emergence from those that do not.

The biomass photosynthesis rate controls the rate at which landscape biomass creates energy via photosynthesis, providing resources for agents over time. The initial biomass site density parameter

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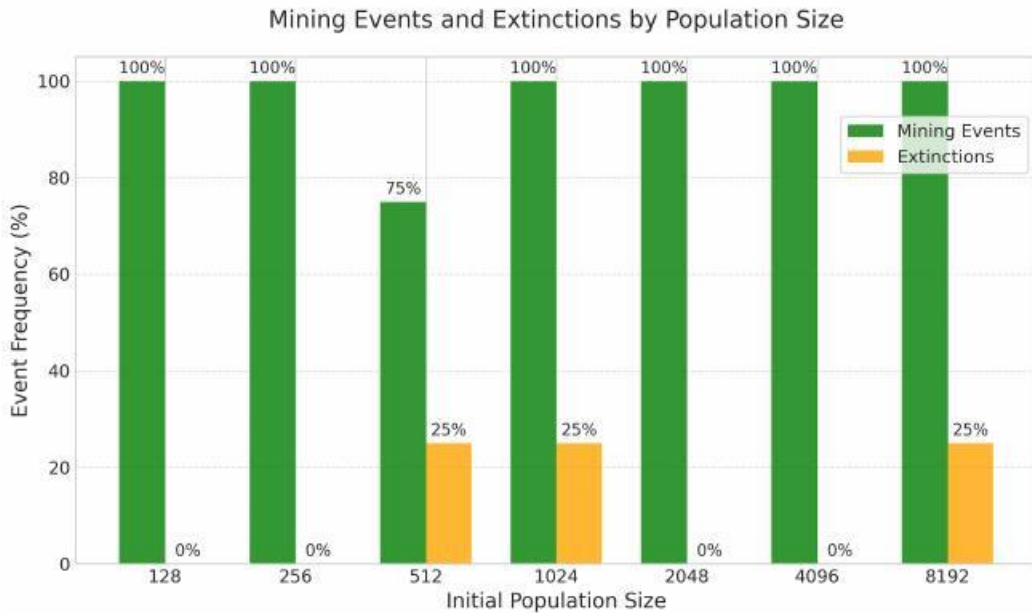


Figure 14: Mining and extinction frequency are flat across initial population size in a 256×256 **Beach** world with (RC) agents (four seeds per setting; 25% extinction means 1/4 seeds reached zero population before 2M steps). Sweeping the starting count over a $64 \times$ range leaves both curves essentially unchanged—isolating initial population as a non-driver, in contrast to the strong world-size effect of Table 1

Table 9: Maximum population reached after step 0 versus starting population size (mean $\pm$ std over four seeds). The peak population is remarkably robust to the starting count: a 128-agent start—just 1.6% of the 8192-agent start—still reaches 78.9% of the latter’s peak (3178 vs. 4026). Larger starts trend to slightly higher peaks with wider seed-to-seed variance, but the effect is far too small to shift mining or extinction outcomes.

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

determines the fraction of grid cells that have biomass at the start of simulation. All experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Whereas initial population size left the outcomes flat (Figure 14), the resource rules move them strongly, as Figures 16 and 17 show—a useful contrast that pinpoints which knobs actually govern emergence.

The mechanism is that larger biomass photosynthesis rates or initial biomass site densities raise the effective carrying capacity by injecting more resources, so larger populations persist, mutations favoring mining become more likely, and extinction becomes rarer. The data bear this out with a clean monotonic trend: averaged over surviving seeds, mean peak population after step 0 climbs

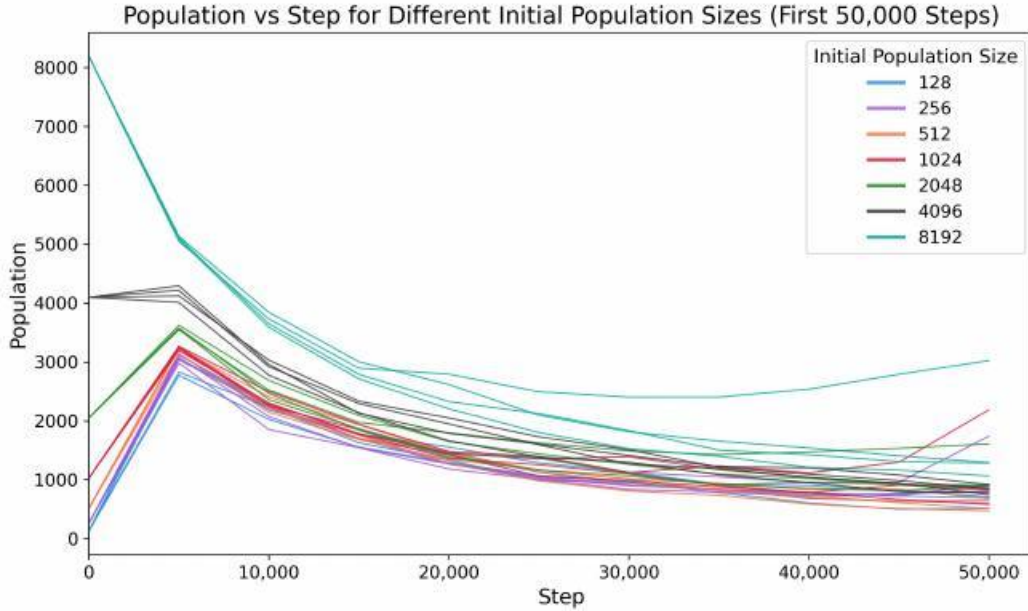


Figure 15: All initial population sizes converge to a common carrying capacity within the first 50,000 steps in the 256×256 **Beach** with (**RC**) agents (four seeds per setting). Curves that start as far apart as 128 and 8192 agents collapse onto nearly the same band, illustrating the resource-limited carrying capacity that makes mining and extinction frequency insensitive to starting size; larger starts raise this capacity only slightly via biomass deposited on death.

from 1608 at initial biomass site density $1/16$ to 2466 at $1/8$, 3786 at $1/4$, and 5353 at $1/2$ —more than tripling as density scales up.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

A natural skeptical hypothesis is that the greater stability of mining in large environments (Table 1) reflects nothing more than self-averaging, rather than qualitatively different dynamics. By self-averaging, we mean the statistical-physics intuition that fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$ with system size N , so that one large system behaves like the average of many smaller ones of the same total size. On this view, a 512×512 **Beach** world would be indistinguishable from many loosely coupled 128×128 worlds run in parallel, and our scaling results could be reproduced simply by running enough small worlds and averaging over seeds. We test this prediction directly and, as shown below, decisively reject it.

To test this hypothesis, we run an experiment holding the total “sample size” roughly fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat this protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

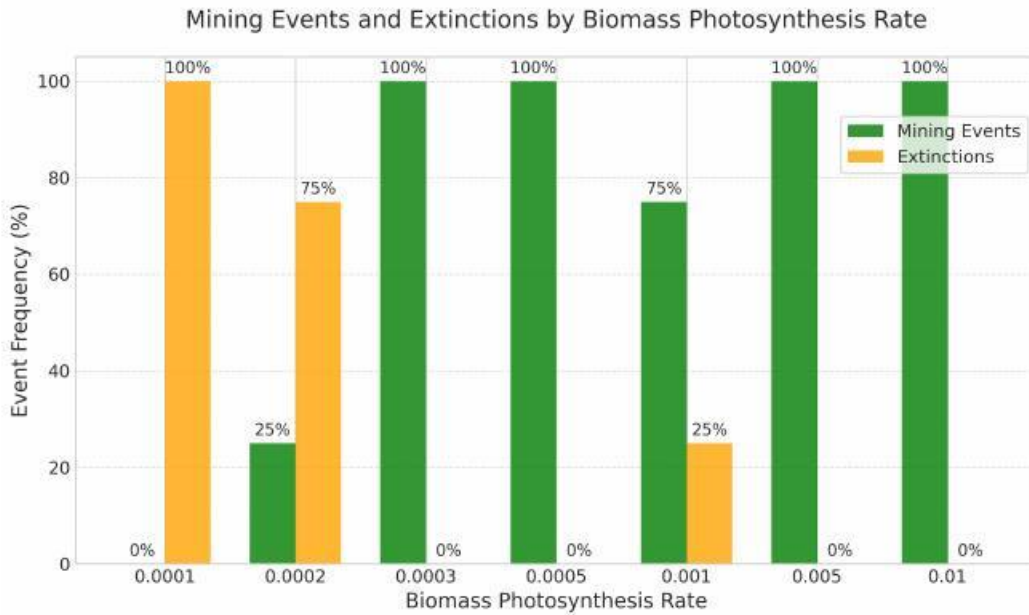


Figure 16: Mining and extinction frequency versus biomass photosynthesis rate in a 256×256 **Beach** world (four seeds per setting). Unlike initial population size, this resource knob has a strong, ordered effect: mining emerges consistently once the rate reaches 0.0003 or higher, and extinction falls off in step.

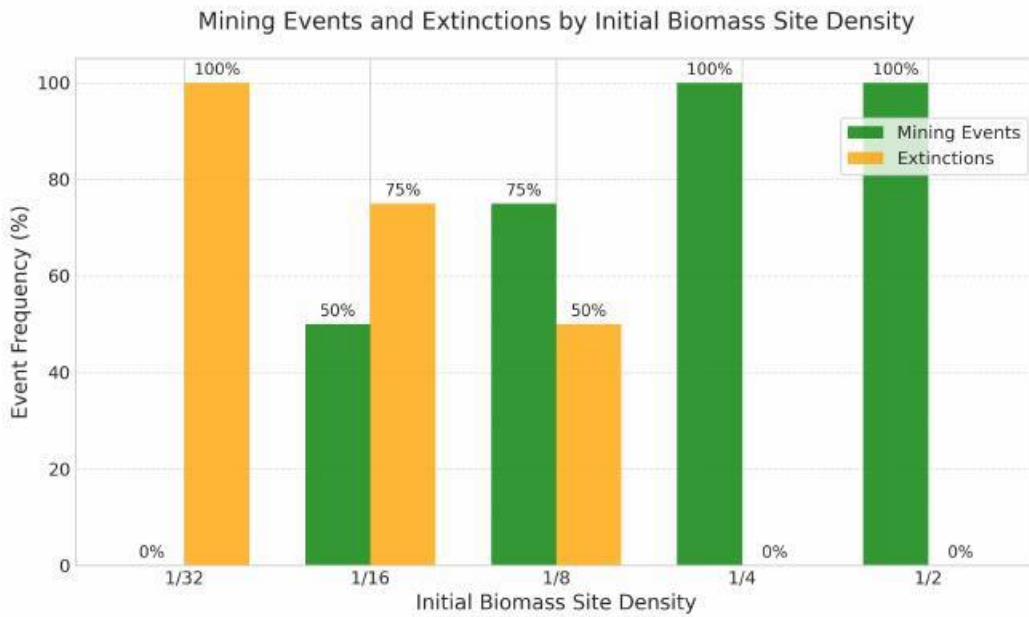


Figure 17: Mining and extinction frequency versus initial biomass site density in a 256×256 **Beach** world (four seeds per setting). The trend is monotonic and matches the peak-population figures in the text: higher densities yield more consistent mining discovery and progressively lower extinction.

Under the self-averaging hypothesis, mining and extinction rates should be roughly invariant to how the fixed budget of grid cells and agents is partitioned—4 large worlds or 64 small ones should discover mining with similar probability. The data show the opposite: outcomes depend strongly on world size. For **(RC)** agents, mining emerges in 4/4 runs with 0/4 extinctions at 512×512 ,

drops to 15/16 mining with 1/16 extinction at 256×256 , and collapses to just 31/64 mining with 41/64 extinctions at 128×128 . The **(R)** control never mines and always goes extinct (0/4, 0/16, 0/64 mining; full extinction at every size), confirming the compass as the key enabler of long-range behavior and ruling out any partition-invariant explanation.

Table 10: Self-averaging cannot explain the scale effect. We hold the total budget of grid cells and initial agents fixed while repartitioning it—4 seeds at 512×512 , 16 at 256×256 , and 64 at 128×128 (initial populations 8192, 2048, 512)—so a self-averaging account predicts identical mining (🔑) and extinction (💀) rates across columns. Instead, for **(RC)** agents mining falls from 4/4 to 15/16 to 31/64 and extinction rises from 0/4 to 1/16 to 41/64 as the budget is dispersed into smaller worlds, while the **(R)** control stays pinned at 0 mining and full extinction throughout.

	512 × 512		256 × 256		128 × 128	
	8192		2048		512	
						
						
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are flatly inconsistent with self-averaging. Rather than smoothing noise, larger environments actively enable and stabilize mining in a way that no amount of averaging over small simulations reproduces.

The comparison is controlled at the level of total cells, agents, and simulation steps, and yet the two allocations land in qualitatively different regimes: concentrating the budget into a few large worlds yields consistent mining and rare extinction (4/4, 0/4 at 512×512), whereas dispersing the identical budget across many small worlds yields populations that frequently fail to mine and often collapse (31/64 mining, 41/64 extinct at 128×128). Increasing environmental scale therefore alters the underlying ecological dynamics—supporting larger, more spatially distributed populations and more persistent policy diversity—rather than merely averaging away stochastic fluctuations.

F.4 MUTATION RATE SWEEP

We test the sensitivity of the emergence of mining to the mutation rate hyperparameter, which controls the scale of Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate directly impacts how different a child agent is from its parent. We once again run experiments in a 256×256 **Beach** world with an initial population of 2048 **(RC)** agents for a maximum of 2M steps.

Figure 18 shows a broad plateau of robustness: across the full order-of-magnitude interval from 6×10^{-3} to 7.875×10^{-2} , mining emerges in at least 3 of 4 seeds, so the mutation rate of 3×10^{-2} used in Section 4.1 sits comfortably in the middle of a stable regime rather than at a fragile optimum. The behavior degrades only at the extremes, exactly as expected: too little mutation (e.g. 3×10^{-3}) starves the population of new behavior and drives it extinct, while too much (e.g. 3×10^{-1}) destabilizes it, again causing consistent extinction with no mining.

F.5 NETWORK SIZE

We study how the size and architecture of the agent policy network affect mining and extinction frequency, and—more importantly—how environmental scale modulates that effect. We compare 256×256 and 512×512 **Beach** worlds (initial populations 2048 and 8192 **(RC)** agents, scaled with grid area). Across every architecture, the **(R)** baseline again sits at the floor, with rare-to-no mining and 100% extinction, so the comparisons below concern only how compass agents respond to network shape at each scale.

The main experiments in Section 4 used 2-layer, 64-channel policies (22,465 parameters per agent; Table 11). Figure 19 shows that, at 256×256 , varying width leaves outcomes essentially unchanged: halving channels to 32 (9,185 parameters) or doubling them to 128 (61,313 parameters) at fixed

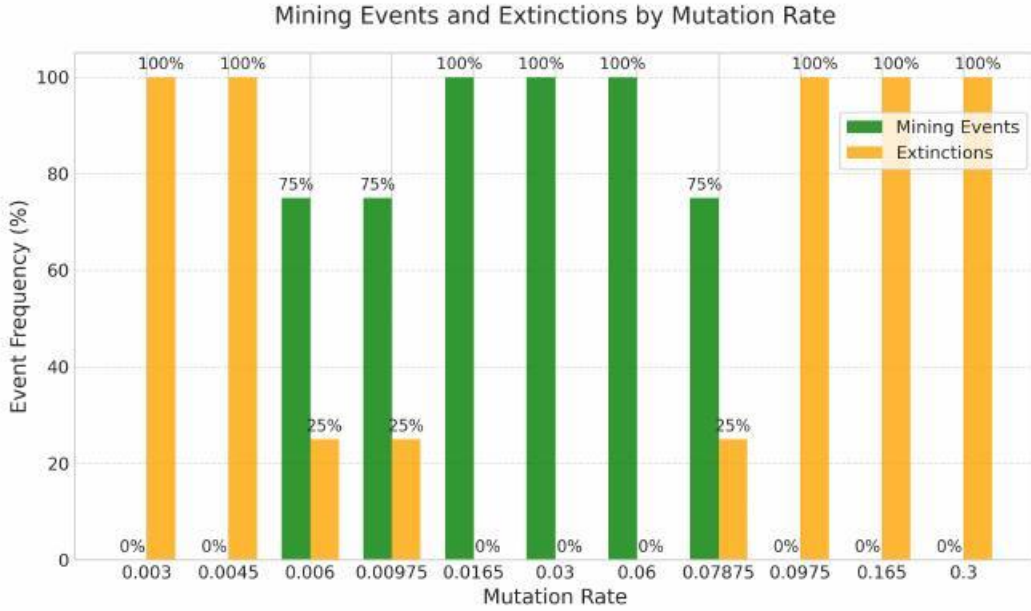


Figure 18: Mining and extinction frequency versus mutation rate in a 256×256 **Beach** world (four seeds per setting). Mining holds at $\geq 3/4$ of seeds across the full 6×10^{-3} – 7.875×10^{-2} interval, with the main-text value 3×10^{-2} centered in this plateau; only the extreme rates collapse to extinction.

depth has minimal to no effect on mining or extinction frequency. Depth, by contrast, is decisive at this scale: all four seeds with a shallow 1-layer, 32-channel network went extinct, as did 50% of seeds with a deep 3-layer, 128-channel network.

Crucially, enlarging the world to 512×512 rescues precisely the depths that failed at 256×256 (Figure 20). In the larger world, the deep 3-layer, 128-channel populations now mine and survive consistently through 2M steps, and—more strikingly—the shallow 1-layer, 32-channel populations do too, reversing the total extinction seen at the smaller scale. This is a second, independent instance of the paper’s central pattern: increasing scale converts a fragile, failure-prone regime into a robust one. We hypothesize that it is more challenging for both smaller and larger networks to discover new behaviors for different reasons. When the network is smaller, the reduced capacity makes it hard to find good solutions. When the network is larger, the increased size of the search space means it takes longer to find promising new behavior. Regardless of the reasons, the fact that these networks fail so often in small environments but succeed in larger ones further illustrates the stability that comes from increasing scale. The larger environment can support larger populations, which allows for more exploration in policy space. Simultaneously, it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For more discussion of these effects see (Mitchell et al. 2006).

Table 11: Total parameters in (RC) and (R) agent policies across the network architectures swept in this section, spanning roughly an order of magnitude from $\sim 8k$ to $\sim 78k$ parameters.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

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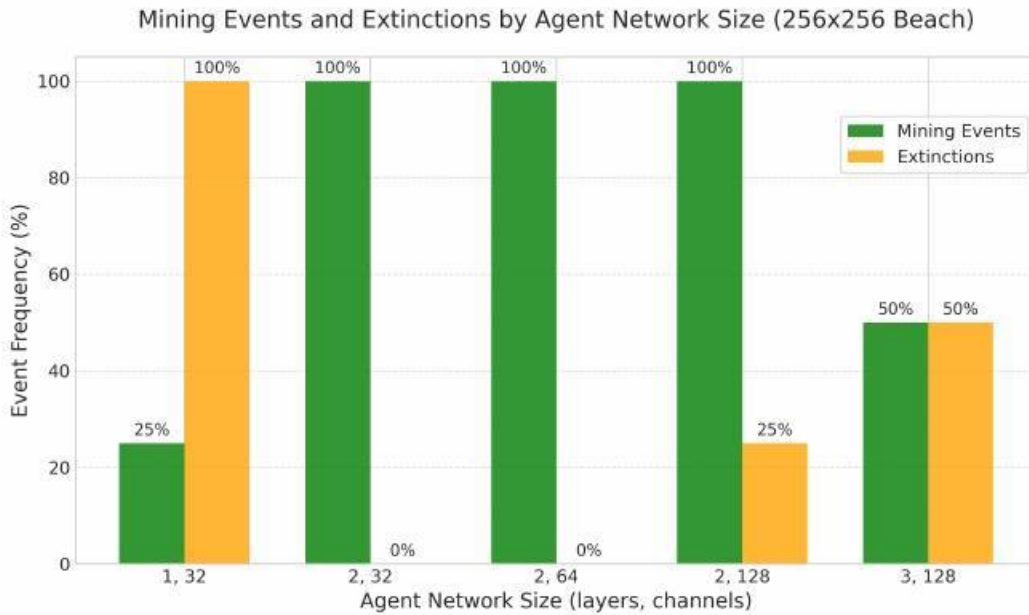


Figure 19: Mining and extinction frequency versus policy network architecture in a 256×256 **Beach** world (four seeds per setting). Width barely matters, but depth is decisive at this small scale: the shallow 1-layer/32-channel network goes extinct in all four seeds and the deep 3-layer/128-channel network in half. Compare Figure 20 where the larger world rescues both.

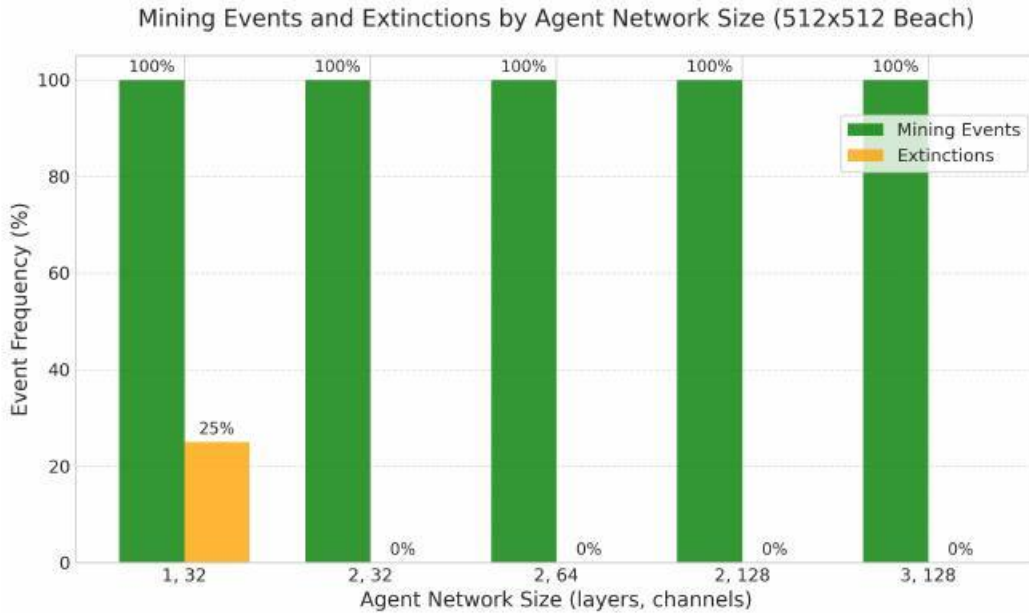


Figure 20: Mining and extinction frequency versus policy network architecture in a 512×512 **Beach** world (four seeds per setting). At this larger scale the depth-driven failures of Figure 19 disappear: both the shallowest and deepest networks now mine and survive consistently, illustrating that increased scale broadens the range of architectures that succeed.

F.6 VARYING VISION RANGE

We measure how vision range and viewing geometry shape foraging and predation for (RCV+A) agents in the 512×512 **Ocean**, sweeping three ranges— 3×3 , 7×7 , and 9×9 —each in two cell-count-matched configurations (Figure 21): a) Around, seeing cells in all directions, and b) Front, seeing no cells behind. The main-text vision experiments (Section 4.2) used the 7×7 Around setting, and we report all metrics relative to a Blind baseline so the marginal value of each additional degree of vision is explicit.

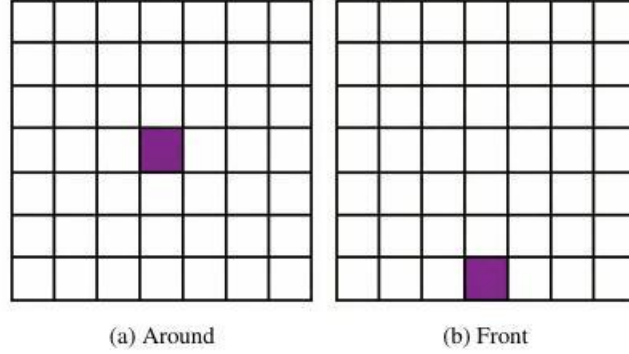


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell represents an agent, facing the top of the page.

Table 12: Foraging and predation versus vision range and viewing configuration for (RCV+A) agents in the 512×512 **Ocean**, with a Blind row for reference (metrics defined in Tables 5 and 6, per-seed averages taken over steps $\geq 500k$ to let the metrics stabilize, then reported as mean $\pm$ std over four seeds). Reading down the range column reveals a saturating curve: 3×3 sits close to Blind, while 7×7 and 9×9 are nearly identical on every metric (e.g. population 16,739 vs. 16,744), evidence of sharply diminishing returns beyond 7×7 .

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

These metrics trace a clear saturating curve in vision range. The smallest 3×3 window behaves much like no vision at all—its Around configuration reaches only a mean population of 10,308 and biomass utilization 0.258, closer to the Blind baseline (7,212; 0.181) than to the larger ranges—whereas 7×7 and 9×9 are nearly indistinguishable from each other (e.g. population 16,739 vs. 16,744 and homicides per attack 0.751 vs. 0.728 in the Around configuration). This is the signature of diminishing returns: the jump from 3×3 to 7×7 is large, but the further jump to 9×9 buys essentially nothing. The pattern is intuitive given that agents are memoryless, so peripheral pixels in a 9×9 window rarely change the optimal action often enough, relative to a 7×7 window, to provide a useful learning signal.

The Around-versus-Front comparison shows a scale-dependent crossover. At the 7×7 and 9×9 ranges, Around is the stronger configuration on every stability and effectiveness metric, posting higher population , biomass utilization , and homicides per attack  than Front (for 7×7 , 16,739 vs. 15,255 population and 0.751 vs. 0.690 homicides per attack). We attribute this to the diminishing value of distant pixels: Around packs more of its budget into nearby cells that carry the information relevant to near-term actions. At the 3×3 range the ordering flips—Front narrowly

beats Around—because when so few cells are visible, one extra cell of forward reach is worth more than one cell behind, and the 3×3 Around setting collapses toward Blind-like behavior.

F.7 VARYING ATTACK STRENGTH

We measure how reducing attack strength reshapes emergent behavior and population dynamics in the 256×256 **Ocean**, comparing three levels: Full Attack (10 HP, an instant kill anywhere in the attack radius), Half Attack (5 HP), and Quarter Attack (2.5 HP). The main **Ocean** experiments in Section 4.2 used Full Attack, which serves as the reference point for the changes reported here.

Halving attack strength from Full to Half Attack cuts both the frequency and effectiveness of violence sharply, with attacks per agent and homicides per attack dropping considerably (Figure 22). Freed from this violence, populations grow larger and agents move and eat more, raising biomass utilization. Cutting strength further to Quarter Attack shows clear diminishing returns, buying little additional reduction in violence or population growth. The mechanism is a compounding “double dip”: weaker attacks require more hits per kill, and because each attack is less lethal, attacking is selected for less strongly—so lower-strength agents both attack less and grow less skilled at it, redirecting their effort toward foraging.

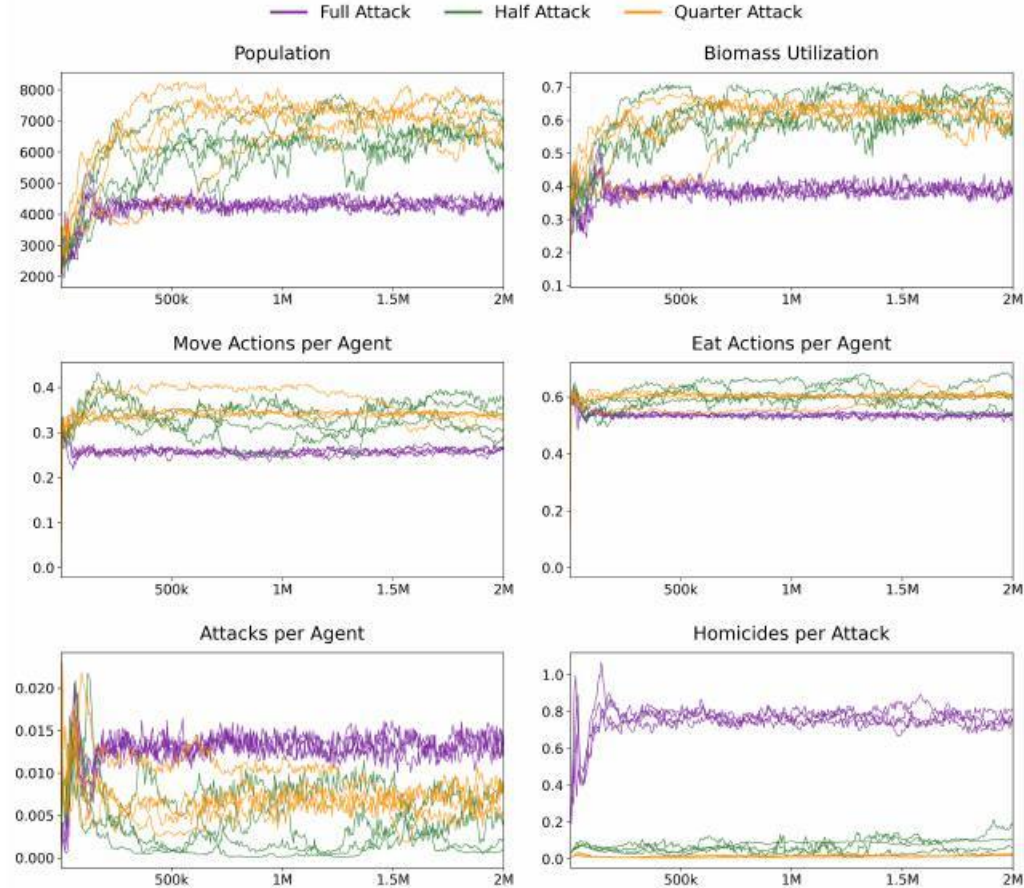


Figure 22: Population and action-frequency trajectories over 2M steps for (RCV+A) agents in a 256×256 **Ocean**, comparing Full, Half, and Quarter Attack strengths (four seeds each). The largest change is the Full→Half step, which sharply lowers attacks per agent and homicides per attack while raising population and biomass utilization; Half→Quarter adds comparatively little, revealing diminishing returns.

F.8 VARYING AGENT HP

We measure the effect of lowering both the starting HP of child agents and the healing rate, which together pressure agents to ‘grow’ over their lifetime in order to survive. The healing rate sets how much HP an agent regenerates per healing step by converting available energy to HP, so agents influence healing only by maintaining enough energy. This ablation asks how much of the emergent-behavior profile shifts when survival is made harder for the newly born.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

The two HP settings are largely matched on homicides per attack, eat frequency, and biomass utilization, which show no meaningful difference (Figure 23). The differences are confined to two metrics: Full HP supports slightly larger populations, while Low HP agents attack somewhat less and move slightly more. We hypothesize that agents born with lower HP must prioritize foraging early in life to reach full size, accounting for this modest shift.

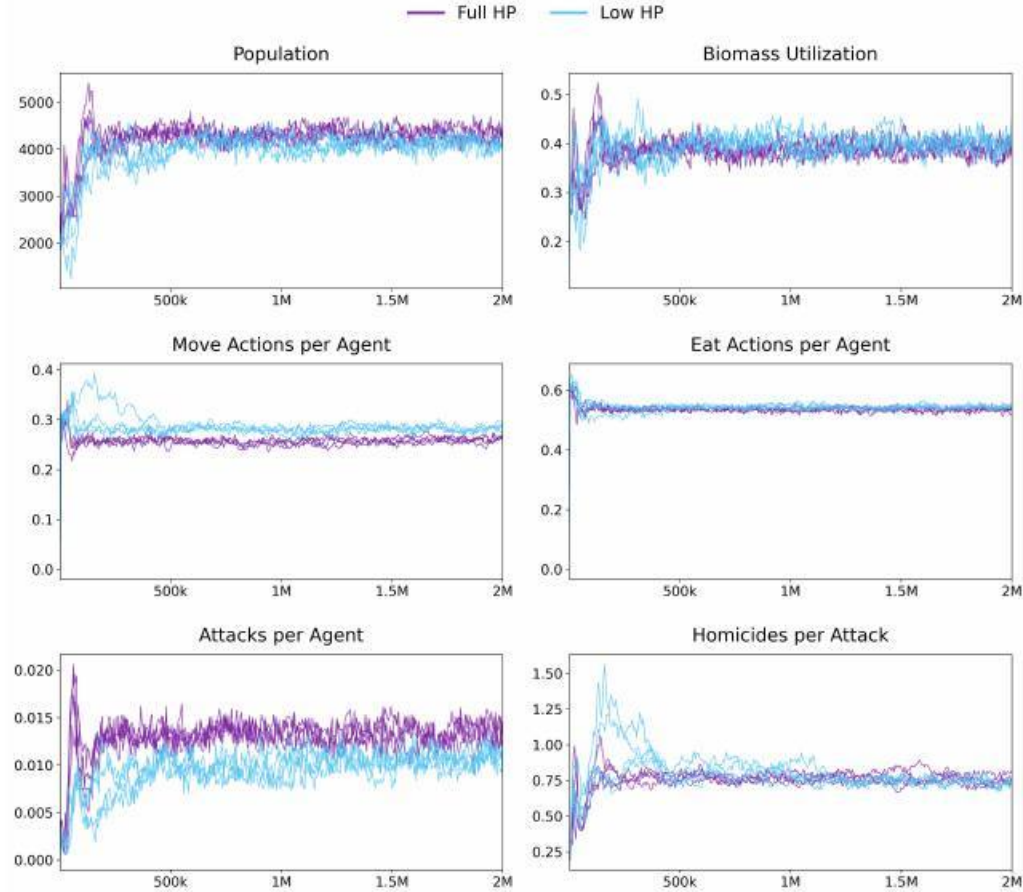


Figure 23: Population and action-frequency trajectories over 2M steps for (RCV+A) agents in the 256×256 Ocean, comparing Full HP and Low HP (four seeds each). The two settings track closely on homicides per attack, eating, and biomass utilization; they diverge only modestly, with Full HP reaching slightly larger populations and Low HP agents attacking a little less and moving a little more.